

Introduction to Human Sciences History Module – Comprehensive Notes

IIIT Hyderabad • Jan 2026

Lectures 1–5 (5 Jan – 22 Jan)

Compulsory Readings

1. E. H. Carr – *What is History?* (Chapters 1–3)
2. Arthur Conan Doyle – *Silver Blaze* (Sherlock Holmes story)

Encouraged / Suggested Readings

1. History of the Temple of Somnath (multiple source perspectives)
2. Umberto Eco – *The Name of the Rose* (novel)

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1 Lecture 1 – Introduction to Human Sciences (5 Jan)

1.1 Course Structure

Module	Discipline	Assessment
1	History	Quiz 1 (45 min)
2	Economics	2*Mid-Sem (90 min, 45+45)
3	Literature	
4	Political Science	
5	Sociology	Quiz 2 (45 min)
6	Philosophy	2*End-Sem (90 min, 45+45)

Grading Philosophy

- Compulsory readings only $\Rightarrow$ B / B+ range.
- Compulsory + encouraged readings + creative thinking $\Rightarrow$ A–/A range.
- Mere reproduction of slides (“photocopying”) $\neq$ A-range.
- Nobody gets full marks in human sciences.
- What is *not* on slides but *spoken in class* **will appear in exams**.

1.2 Why Human Sciences at an Engineering College?

Historical Context of Engineering Education

- **MIT model:** humanities integrated into engineering education.
- **IIT history:** derived from older institutions (e.g. Thomason College of Civil Engineering, Roorkee – one of the earliest engineering colleges in Asia).
- Engineering does not exist in isolation; engineers must understand the society they work in.

1.3 Anglophone vs. Francophone Distinction

Key Terminological Distinction

- **Anglophone (English-speaking world):** “Humanities” and “Social Sciences” are separate categories.
- **Francophone (French-speaking world):** Uses the umbrella term “**Sciences Humaines**” (Human Sciences) – encompasses both.
- **Human Sciences $\neq$ Humanities $\neq$ Social Sciences:** Human Sciences is a broader, unified field studying the human condition.

1.4 Pre-Modern Knowledge: Divine Revelation

Sources of Knowledge in Pre-Modern Societies

Before modern science, knowledge came primarily through **divine revelation**:

- **Vedas / Vyasa:** knowledge revealed through meditation and divine insight.
- **Moses:** Ten Commandments received from God.
- **Prophet Muhammad:** Quran revealed by God through Angel Gabriel.
- **Buddha:** Enlightenment through meditation under the Bodhi tree.

Key characteristic: Knowledge was gained through meditation / divine intervention, *not* through empirical observation or experimentation.

Subject of knowledge: God / the divine (not the human being).

Theology was the queen of all sciences.

1.5 The Heliocentric Revolution

The Great Shift: De-centering of Creation

Copernicus (1543) → **Galileo** (early 1600s):

- Earth is *not* the centre of the universe; the Sun is.
- Galileo's telescope: observed Jupiter's moons ⇒ used to **measure time** precisely.
- Laws of nature are **universal**: same for a king and a peasant, a Brahmin and a Dalit.
- No special dispensation based on social status, religion, or birth.

Newton took this further:

- **Absolute laws of nature** govern everything.
- God reduced to the role of “**First Cause**” only – set the universe in motion, then natural laws take over.
- No miracles, no divine intervention in daily affairs.

Disenchantment of Nature

The idea that nature operates by **fixed, discoverable laws**, not by divine whim or miracles. Nature is “disenchanted” – stripped of magical or supernatural explanations.

Example: Birsa Munda Revolt

Birsa Munda led a tribal revolt against the British using **archaic/traditional knowledge** (bows and arrows, belief in divine protection). The British had **guns** – products of the new science. The revolt was crushed. This illustrates the clash between pre-modern and modern knowledge systems.

1.6 Renaissance and Humanism

Renaissance Shifts

- **Human form in art**: Leonardo da Vinci's anatomical studies; perspective in painting.
- Shift from **predestination / divine will** to **human agency**.
- The **human being** becomes the central subject of knowledge.

Humanism (Academic Meaning)

Humanism: A perspective in which **human beings are centred** and **human actions are analysed**. It does *not* mean “being humane” or “humanitarian” in everyday usage. It means studying humans *as* the subject of inquiry.

1.7 De-centering of Creation & De-centering of Human Beings

Two Great De-centerings

1. **De-centering of Creation**: Earth is not the centre (Copernicus/Galileo).
2. **De-centering of Human Beings**: Humans are not special creations – subject to the same natural laws as everything else.

Carl Sagan's *Pale Blue Dot*: Earth as a tiny speck in the cosmos ⇒ ultimate de-centering.

1.8 Human Sciences vs. Natural Sciences

Fundamental Difference

- **Natural Sciences:** Finite causes → predictable effects (e.g. gravity).
- **Human Sciences:** **Infinite antecedent causes** – every human action is the product of countless prior causes (upbringing, culture, economics, psychology...).
- You *cannot* isolate variables in human society the way you can in a physics lab.
- The **foundation of knowledge** is fundamentally different.

1.9 Emergence of Modern Disciplines

Discipline	Approx. Origin	Focus
History	First to emerge; dept. ~1810 (Copenhagen/Berlin)	Study of the past; theory of change
Economics	1780s (Adam Smith, <i>Wealth of Nations</i>)	Production, distribution, consumption
Sociology	1850s–60s (Industrial Revolution)	Social relations and organisation
Political Science	19th century	Governance and power
Literature	Part of humanities tradition	Cultural expression and themes
Philosophy	Ancient; formalised later	Ethics, knowledge, existence

Interdisciplinarity Example: Coal Mining

A colleague studies coal mining communities in Chhattisgarh and Telangana. In Chhattisgarh, there is widespread “theft” of coal by locals; in Telangana, it does not happen. **Same type of mine, different social outcome.** Each discipline asks different questions:

- **Sociologist:** What social structures differ?
- **Economist:** What are the economic incentives?
- **Political Scientist:** How do governance structures differ?
- **Historian:** What is the history behind these communities and mines?
- **Philosopher:** What are the ethical dimensions?

Another Example: Airplane Behaviour

Singapore Airlines flight: passengers board quietly in Singapore, but on landing in Hyderabad, chaos erupts – phones ringing, bags opening before the plane stops. Same people, same flight: what changed? Sociology, philosophy of ethics, and cultural studies each offer a different lens.

2 Lecture 2 – History as Academic Discipline: Facts & Evidence (8 Jan)

2.1 What is History?

History = Theory of Change

History is the study of **human and natural causes** of past events. It is a **theory of change**: to identify change, you must first identify what is **not changing** (constants).

E. H. Carr – *What is History?* (Compulsory Reading)

- History is a **continuous process of interaction** between the historian and their facts.
- Facts do not speak for themselves – they are *selected* and *interpreted* by historians.
- “Study the historian before you study the facts.”

Key Principles

- History is everywhere – everything has a past, present, and future.
- Academic history has **protocols** distinct from school/popular history.
- There is **no certainty** in history – only **levels of possibility** and **degrees of probability**.
- The **Sherlock Holmes analogy**: like a detective, the historian builds a picture from scattered, incomplete evidence.

2.2 The Concept of “Fact” in History

Traditional Definition of Fact

A “fact” was traditionally seen as something **objective**, existing outside interpretation – like legal evidence in a court. But the history of the concept shows this is more complex.

2.3 Evolution of Historical Method

Figure	Period	Contribution
Herodotus	5th c. BCE	First historian; “historia” = inquiry. Collected evidence about Greco-Persian Wars.
Thucydides	5th c. BCE	Evaluated evidence; neutral observer of Peloponnesian Wars (Athens vs Sparta). Origin of the “objective fact.”
~1500 years of revealed / divine history		
Ibn Khaldun	14th c. CE (Tunisia/Algeria)	Developed principles to evaluate evidence ; part of Islamic Renaissance.
Ibn Rushd (Averroes)	12th c. CE	Categorised errors in historical investigation : partisanship, unexamined beliefs, insufficient knowledge.
Al-Khwarizmi	9th c. CE (Khorasan)	Origin of the word “algorithm”; algebra – same intellectual milieu.
Edward Gibbon	18th c. CE	Distinguished primary vs. secondary sources ; wrote secular history of Christianity (<i>Decline and Fall of the Roman Empire</i>).
Leopold von Ranke	19th c. CE	Historians should work only with facts, no interpretation ; “wie es eigentlich gewesen” (as it actually was). Wrote history of Catholic Popes <i>as a Protestant German</i> .

Key Insight: Facts Are Constituted, Not Found

Facts are **constituted by human beings**, not purely objective entities waiting to be discovered. The historian *selects* which facts matter, *interprets* them, and *arranges* them into a narrative.

2.4 Verifiability and Falsifiability

Standards of Historical Knowledge

- Historical claims must be **verifiable**: based on **open, accessible** evidence.
- They must also be **falsifiable**: it must be *possible* to prove them wrong.
- Without these standards, history degenerates into myth, propaganda, or fiction.

Example: Jallianwala Bagh Massacre

How many people were killed? This seems like a simple factual question, but:

- **Primary evidence**: number of bullets fired, bullet holes in walls.
- **Estimation**: body counts, eyewitness testimonies (which vary widely).
- The “fact” of the death toll has **changed over time** as new evidence emerged and old estimates were revised.

Archaeology Example: European Caves

Caves in Southern Europe, 500,000–400,000 years ago: evidence of fire use by humanoids. How do we know? Charcoal residue, blackened walls, arrangement of stones. The “facts” are reconstructed from scattered physical evidence.

Facts Change!

- Facts can **change importance** over time.
- **New facts emerge** as new evidence or methods of analysis appear.
- **Old facts get disproven** by better evidence.

3 Lecture 3 – Historical Time & Periodisation (12 Jan)

3.1 History vs. the Past

Crucial Distinction

- **The Past** is amorphous, infinite, undifferentiated – everything that has ever happened.
- **History** is a specific, **human-manufactured narrative** with a **beginning and an end**. It is curated, selected, and interpreted.
- History $\neq$ Past.

3.2 Prehistory vs. History

Traditional Divide

Prehistory: The period *before* civilisation. Civilisation requires three markers:

1. **Settled life** (not nomadic)
2. **Writing / script**
3. **Cities / states**

Questioning the Divide

The **Rig Vedic people** complicate this:

- They had complex social organisation and arguably proto-states.
- But they were **mobile** (not fully settled) and had **no writing** (oral tradition only).
- Are they “prehistoric” or “historic”? The categories are not clean.

3.3 Conceptions of Time

Model	Tradition	Description
Cyclical	Indian (Yugas: Satya → Treta → Dvapara → Kali → repeat)	Time repeats in endless cycles; no linear progress.
Spiral	Aztec / Mesoamerican	Time spirals – repeats but with variation.
Arrow (Linear)	Western / Abrahamic	Time moves in one direction: past → present → future. Used in academic history.

Dating Conventions

- **BC/AD** (Before Christ / Anno Domini) → now replaced by **BCE/CE** (Before Common Era / Common Era) in academic usage.
- **c.** = *circa* (approximate date).

3.4 Stone Ages

Period	Dates (approx.)	Characteristics
Paleolithic	2.5 million – 10,000 BCE	Stone tools; mobile hunter-gatherer life.
Mesolithic	10,000 – 5,000 BCE	Mortar and pestle; bone needles (still in use today).
Neolithic	5,000 – 2,000 BCE	Agriculture begins; settled life emerges.

3.5 Metal Ages

Period	Dates (approx.)	Characteristics
Chalcolithic	~3,000–1,500 BCE	Copper + stone used together.
Bronze Age	~1,500–500 BCE	Alloys (copper + tin); more complex tools and weapons.
Iron Age	~500 BCE – 500 CE	Large empires; organised religion (Buddhism, Jainism); Mauryan Empire.

Technologies Overlap!

The Iron Age has never truly ended – we still use iron. Technologies do not neatly replace each other; they **overlap and coexist**. Bone needles from the Mesolithic are still used today.

3.6 Problems with Periodisation

Why Periodisation is Problematic

1. **Moral loading:** “Ancient” = golden age; “Medieval” = dark ages; “Modern” = progress. These are **value judgments**, not neutral categories.
2. **Eurocentrism:** Ancient/Medieval/Modern was designed for European history. It does *not* fit non-Western histories well.
3. **When does “modern” begin in India?**
 - 1757 – Battle of Plassey?
 - 1793 – Permanent Settlement?
 - Arrival of railways?
 - Establishment of universities?
4. **Different regions** entered periods at different times.
5. **Technology vs. social organisation:** which should define a period?

3.7 Key Historical Perspective Points

Time Scale Perspectives

- **Indus Valley Civilisation** (~2500–1500 BCE): not one civilisation but many changes over **2,000 years**. Language, religion, social structure could not have remained constant.
- **Ashoka** is closer to us in time than to the establishment of Mohenjo-daro.
- **Satavahana dynasty:** survived ~500 years.
- **Kushan Empire:** ruled ~300 years; territory from Uzbekistan to Uttar Pradesh.

4 Lecture 4 – Indian Historiography: Schools of Thought (19 Jan)

4.1 Traditional Indian Historiography

Shruti and Smriti

- **Shruti** (“that which is heard”): oral traditions, Vedas – considered divinely revealed.
- **Smriti** (“that which is remembered”): written traditions, epics, Puranas, law codes.
- Time is **cyclical**; no distinction between past/present/future in the Western linear sense.
- Actors are often **divine** or semi-divine.
- Family genealogies linked to **Suryavanshi** (Solar) and **Chandravanshi** (Lunar) dynasties.

Lokayata / Charvaka Tradition

An ancient Indian **materialist**, **empiricist** philosophical tradition. Rejected divine revelation; argued for evidence-based knowledge. **Sources were destroyed** – we know about them *only from hostile references* by their opponents (Buddhist, Jain, Brahmanical texts).

4.2 British Engagement with Indian History

4.2.1 Phase 1: Orientalists (late 18th – early 19th century)

British Orientalists

- **Admired** Indian culture, languages, and learning.
- **William Jones**: founded the **Asiatic Society of Bengal** (1784).
- Studied Sanskrit; drew connections between Sanskrit, Greek, and Latin (Indo-European language family).
- Approach: learn Indian languages, study Indian texts, translate them.

4.2.2 Phase 2: Utilitarians / Liberals (1820s onward)

The Utilitarian Shift

- **James Mill**: wrote *History of British India* (6 volumes) – **never visited India**.
- Introduced the **Hindu / Muslim / British** periodisation of Indian history (still influential and problematic).
- **Thomas Babington Macaulay**:
 - Changed education policy to **English + Western sciences**.
 - Drafted the **Indian Penal Code**.
 - Famous “Minute on Education” (1835).
- View: Indians are “backward” and “savage”; British have a civilising mission.
- “**White man’s burden**” (Rudyard Kipling) – ideology of colonial responsibility.
- **Religion, Language, and Caste** became the primary categories to understand Indians.

4.3 Key Archaeological Milestones

Figure / Event	Contribution
James Prinsep	Deciphered Brahmi script (1830s, Banaras) – unlocked ancient Indian inscriptions.
Alexander Cunningham	Established the Archaeological Survey of India (ASI) in 1861; discovered Ashoka’s edicts.
Harappa / Mohenjo-daro discovery (1920s)	Pushed Indian history back by ~3,000 years ; revealed a sophisticated urban civilisation.

4.4 Four Schools of Indian Historiography

School 1: British Colonial Historiography

- India had **no political unity** before the British.
- **Oriental despotism**: all Indian rulers were tyrants.
- **Unchanging village communities**: India was static, frozen in time.
- The **British as the agent of change** and modernisation.
- James Mill's Hindu/Muslim/British periodisation.

School 2: Nationalist Historiography

- **India did have unity**: Mauryan, Kushan, Mughal, Maratha empires.
- **Unifying ideologies**: Buddhism, Bhakti movement, shared cultural traditions.
- Key figures: **Jawaharlal Nehru** (*Discovery of India*); **Romila Thapar** (on Ashoka).
- **R. C. Dutt**: economic history showing the “**drain of wealth**” from India to Britain.
- **Jadunath Sarkar**: major historian of the Mughal and Maratha periods.
- Purpose: counter the colonial narrative; build national identity.

School 3: Marxist Historiography

- **D. D. Kosambi**: mathematician at TIFR; pioneer of Marxist history in India.
 - Focused on **material foundations** of history.
 - Argued: **iron technology** enabled the Mauryan Empire.
- **R. S. Sharma**: studied material culture and class structures.
- **Irfan Habib**: studied the **Mughal agrarian crisis** – how exploitation of peasants led to empire's decline.
- Focus on **class struggle** and economic base of society.
- View: **1947 was not a fundamental change** – the same class structures continued.

School 4: Subaltern Studies (1981, Calcutta)

- Founded in 1981 in Calcutta; spread globally (Latin America, Africa).
- Focus: the **indigenous worldview** of peasants, tribals, and marginalised people.
- Critique of **both** Western colonial **and** elite Indian (nationalist/Marxist) perspectives.
- **Birsa Munda** example: his revolt was not “pre-political” – it had its own logic.
- Concept of **epistemic violence**: imposing Western categories of knowledge on non-Western peoples *is itself* a form of violence.
- **Critiques of Subaltern Studies**:
 - Who exactly is “subaltern”? The category is slippery.
 - Risk of **problematic essentialism** – romanticising the subaltern.
 - Can the subaltern truly “speak” through academic texts written by elites?

4.5 Comparison Table: Four Schools

School	Core Argument	Key Figures	On 1947
Colonial	No unity; British = modernisers	James Mill, Macaulay	British created India
Nationalist	India had unity & civilisation	Nehru, R. C. Dutt, Thapar	Liberation; nation restored
Marxist	Material base; class struggle	Kosambi, R. S. Sharma, Irfan Habib	Not fundamental change
Subaltern	Voice of the marginalised	Ranajit Guha and collective	Elite transfer of power

4.6 Age of Consent Controversy (1890s)

Age of Consent Controversy

In the 1890s, debates over raising the age of consent for girls revealed deep tensions between:

- **Reformers** who wanted to modernise Indian society.
- **Cultural nationalists** who saw British-imposed reform as an attack on cultural sovereignty.
- Even those who agreed on the goal disagreed on *who* should implement it (Indians themselves vs. British law).

4.7 New Methodologies in History

Emerging Methods

- **Computational tools:** NLP (Natural Language Processing) for analysing texts.
- **GIS** (Geographic Information Systems): mapping historical data spatially.
- **Oral histories:** recording the voices of people who left no written records.
- **Environmental history:** studying the relationship between humans and their environment over time.

5 Lecture 5 – Historical Geography of India (22 Jan)

5.1 Rethinking Indian History

Beyond Empires and Kings

Typical Indian history = succession of empires, kings, religions, battles. But historians are now saying this is **grossly insufficient**. We need to understand:

1. River basins
2. Fields (agriculture)
3. Forests
4. Mountains

...and their interplay to truly understand Indian history.

5.2 Empires That Don't Fit the Standard Map

The Kushan Empire (~300 years)

Territory: from Uzbekistan, Tajikistan, Turkmenistan, parts of Iran, Afghanistan, Pakistan, Kashmir, Punjab, Haryana, Rajasthan, parts of MP, Bihar.

For **15–17 generations**, people from these diverse regions lived under a single political entity. They could travel freely, paid the same taxes, were governed by the same laws. Yet Indian history barely discusses this because it doesn't "fit the map" of modern India.

Rajendra Chola's Empire (~200 years)

- Capital: Gangaikondacholapuram ("City of the Chola who conquered the Ganges").
- Conquered: Kalinga, Pala rulers of Bengal/Bangladesh, and extended influence to Burma, Myanmar, Cambodia, Thailand, Sumatra, Bali.
- From Hyderabad to Southeast Asia in the same "country."
- He went *east and south*, not north – an aberration from the usual template.
- **No one did this before or after him** in Indian history.

5.3 River Basins as Historical Entities

Rivers: The Pillars of Indian History

River basins are not just geographical features – they are **historical entities** that sustain life and are self-contained units.

- **Indus Valley**: sustained human habitation continuously for ~2,000 years.
- **Ganges Valley**: sustained continuous life for ~2,500 years and counting.
- Other major basins: Godavari, Krishna, Kaveri, Mahanadi, Narmada.

5.3.1 Indus vs. Ganges: Crucial Differences

Indus Valley	Ganges Valley
Flows through desert / desert-like hinterland.	Extremely moist, alluvial soil.
Agriculture only right next to the river (like the Nile).	Agriculture across vast plains; dig 10 feet ⇒ water even without rain.
Limited agricultural surplus.	Massive agricultural surplus ⇒ cities, empires.
Habitations more prone to disruption.	Continuous habitation for 2,500+ years.
Gets ~7 weeks less monsoon rain.	Full monsoon season.

The Ganges Valley = “Powerhouse of Indian History”

Only **two** geographical features on Earth have comparable agricultural potential:

1. The **Ganges Basin** (India)
2. The **Yangtze Basin** (China)

India’s historical prosperity is not due to any culture, race, or religion – it is because of this **exceptional geography**: massive surplus $\Rightarrow$ feed soldiers, artisans, builders, scholars $\Rightarrow$ civilisation.

5.4 Mountains, Forests, Fields: Enabling and Limiting

Dialectics

Dialectics: Reality created by the **coming together of contradictory forces**. E.g., movement and stability; forest and field; conflict and complementarity. Every enabling condition is *also* a limiting condition.

Mountains

- **Enabling**: protection from invaders, winds, climate regulation.
- **Limiting**: no large-scale agriculture $\Rightarrow$ no large states/empires.
- Mountain peoples (tribals): Santhals, Bhils, Gonds, Todas, peoples of Arunachal/Meghalaya.
- States never fully controlled mountains – from Ashoka to Aurangzeb to post-independence India (Kashmir, Nagaland insurgencies).
- This is **not unique to India**: Burma, China, Pakistan (Pashtuns) – same pattern globally.

Forest vs. Field: The Central Dialectic of Indian History

- **Conflict**: agriculture requires clearing forests; industrial development destroys forests (mining, roads, cities).
- **Complementarity**: forests provide charcoal (essential for iron-making), grazing, biodiversity, soil renewal.
- Animals from agricultural areas grazed in forests, left manure, created biomes.
- Rivers: both **life-sustaining** and **destructive** (floods kill cattle and people, but deposit alluvial soil for bumper harvests).
- This conflict/complementarity **continues today**: tribal displacement, mining, deforestation.

5.5 The Deforestation of the Ganges Valley

A 2,000-Year Human Artifact

- **3,000 years ago**: the Ganges plain from Delhi to Dhaka was **impenetrable tropical forest**, at least 15,000 years old.
- Over **2,000+ years**, humans systematically deforested it using **iron tools** (axes).
- **UP today**: $\sim 2\%$ forest cover (from 100%).
- This flat agricultural landscape is a **human artifact** – as much a creation as any building or textile.
- It was a **massive collective effort** by people who were not centrally organised.
- Deforestation **changed everything**: monsoon patterns, soil, flora, fauna.

5.6 Horse, Fire, and Iron: The Three Keys

The Triad That Made Indian History

Control over the horse → **control over fire** → **control over iron**. In that order.

1. **Horse**: essential for domesticating animals at scale (cowboys; herding hundreds of cows, goats, sheep).
2. **Fire**: turning iron ore (ferrous oxides) into usable iron requires **charcoal** (wood burned without oxygen).
3. **Iron**: axes to cut forests ⇒ agriculture ⇒ surplus ⇒ empires.

Without mastery of the horse, you cannot master large-scale agriculture.

Without fire (charcoal), you cannot make iron.

Without iron, you cannot deforest the Ganges valley and create surplus.

5.7 Agriculture and Domestication

Key Definitions

- **Agriculture**: domestication of **plants** – specifically **grass**. Rice and wheat are domesticated grasses, bred over hundreds of thousands of years.
- **Pastoralism**: domestication of **animals** – cows, goats, sheep, horses, dogs.
- Settled agriculture is only ~2,500 years old; agriculture itself is ~10,000–12,000 years old.
- Bhimbetka caves: among the earliest evidence of agriculture in India.

5.8 New Environmental Conditions ⇒ New Everything

Chain Reaction of Deforestation

1. Iron technology → forest clearing.
2. New environmental conditions: new types of rice, vegetables, crops.
3. New settlement patterns → new social organisation.
4. 300–400 years after serious deforestation: **Buddhism and Jainism** emerge.
5. Jainism's impact on Vedic religion: shift from **animal sacrifice** to **non-violence** (ahimsa).
6. New economies and politics ⇒ Mauryan Empire, Gupta Empire, etc.
7. **If the Ganges valley had not been deforested, Indian history would have been entirely different.**

5.9 Energy and Historical Change

Energy Source	Calories returned per 100 spent	Historical Impact
Human labour	~105–106	Baseline; limited surplus.
Horse	~112–114	Enabled large-scale herding and agriculture.
Charcoal	~125	Enabled iron-making; constant for 2,000 years.
Mineral coal	~187–190	Revolutionary ; changed everything.

Mineral Coal Changed History

- Until the 19th century, charcoal was the energy source – it came from trees (deforestation!).
- The discovery of **mineral coal** in the late 1800s gave ~190 calories return per 100 spent.
- **This is why the British won.** Not because of racial superiority, not because of ideas of liberty – because they had a **new form of energy**.
- Mineral coal enabled: industrial manufacturing, railways, steamships, modern weapons.

5.10 The Ganges Valley as Power Centre

2,000 Years of Power from the Ganges

- From the start of major deforestation, the Ganges plains became the **power centre of India**.
- Every major empire needed to control the Ganges: Ashoka (Pataliputra), Shah Jahan and Aurangzeb (Agra/Delhi).
- **Muhammad bin Tughlaq**: shifted capital from Delhi to Daulatabad (Deccan) – scientifically rational (centre of India) but the empire **collapsed** because he left the Ganges.
- **Aurangzeb**: shifted focus from Ganges to Deccan to fight Marathas ⇒ Mughal decline.
- **Post-independence**: most popular PMs (Nehru – Kashmiri; Modi – Gujarati) still contest elections from Ganges valley constituencies (Allahabad, Varanasi).
- Only **one** PM of consequence was not from the Ganges constituencies.

5.11 British Contributions to Historical Geography

5.11.1 Railways

Railways Changed Everything

- The **first empire from the sea coast** since the Cholas.
- Railways connected geographies that had **never been in regular contact**: Punjab to Bombay.
- Enabled seasonal migration (UP/Bihar workers to Bombay during non-monsoon months).
- Cultural exchange: Tamil Nadu weddings now have Punjabi *ladies' sangeet* (via Bollywood, enabled by railways → cultural mixing).
- **Shiv Sena** politics in Maharashtra: a product of migration patterns enabled by railways.

5.11.2 The Census

The Census: Creating a New India

Timeline:

- **1824**: first attempt in Ahmedabad.
- **1827**: Banaras (where James Prinsep was deciphering Brahmi script).
- **1856**: general census ordered, but postponed due to **1857 Rebellion**.
- **1871–72**: first modern census (non-synchronous; took 6 months; registered every habitation).
- **1881**: **first synchronous census** – population of all India counted on the midnight of 17 February. One schedule of questions for *everyone*. **Gave a picture of India that had never existed before.**
- **1931**: most extensive demographic record; still a valid reference document today.

Key innovation: every *Homo sapiens* counted, regardless of caste, gender, religion, age.

Problem: census used **religion, caste, and language** as *primary* categories (unlike Europe, which used occupation, gender, location). This:

- Made populations **self-aware** of their numbers.
- Enabled **political mobilisation** based on community identity.
- Created new identity categories and solidified existing ones.

Vande Mataram and the Census

Bankim Chandra Chattopadhyay wrote *Vande Mataram* for his novel *Anandamath*. The original line: “*Saptakoti kanthe kala-kala*” (“seven crore voices rose in unison”) – the figure of **seven crore** comes from the **1881 census** which showed Bengal’s population as seven crores. The national song itself was shaped by census data!

5.11.3 Trigonometric Surveys and Maps

The Great Trigonometrical Survey

- Started **1802** after defeating Tipu Sultan (1799).
- First base: **Anna Salai** (Mount Road), Madras, near Fort St. George.
- Covered **all of India** by **1872**.
- Used **trigonometry on a curved surface** (unprecedented at scale).
- **Mt. Everest**: calculated in the 1850s from the **south bank of the Ganges at Patna** – a tiny change in the horizon, measured using trigonometric formulas.
- The first reliable map of India was finalised around **1890** – barely 100+ years ago.

Before Modern Maps: The Jambudvipa Mandala

Before trigonometric maps, Indians used **mandala maps**:

- Central circle = India; centre = Mount Meru.
- Actual rivers and kingdoms shown symbolically.
- Concentric circles: seas, distant lands, realms of gods.
- Rajendra Chola, Kanishka, Harsha **actually used these maps** to navigate and plan military campaigns.
- The word “Koromandel” (Coromandel Coast) comes from these mandalas.
- **300 years ago, this was the operational map of India.** Nobody today can read it.
- Historical orientation was to the **east** (where the sun rises) – hence the word “orient” / “orientation.”

Geographical Determinism – A Critique

Some scholars argue geography “determines” everything. But the professor emphasised: geography provides **enabling and limiting conditions**, not deterministic outcomes. It took **human agency** (horse, fire, iron) to convert the Ganges valley’s potential into actual power.

6 Master Summary: All Five Lectures

Lecture	Core Theme	Key Takeaways
1 (5 Jan)	Introduction to Human Sciences	Heliocentric revolution; disenchantment of nature; humanism; human $\neq$ natural sciences; infinite antecedent causes.
2 (8 Jan)	Facts & Evidence in History	E. H. Carr; facts are constituted; verifiability & falsifiability; evolution from Herodotus to Ranke.
3 (12 Jan)	Historical Time & Periodisation	History $\neq$ past; cyclical vs linear time; Stone/Metal Ages; periodisation is Eurocentric and morally loaded.
4 (19 Jan)	Indian Historiography	Four schools: Colonial, Nationalist, Marxist, Subaltern; British Orientalists to Utilitarians; Prinsep, ASI, IVC.
5 (22 Jan)	Historical Geography	River basins; Ganges as powerhouse; forest vs field; horse-fire-iron triad; coal; census; trigonometric surveys.

7 Glossary of Key Terms

Term	Definition
Disenchantment of Nature	Nature operates by fixed laws, not divine whim or miracles.
Humanism	Perspective centring human beings and analysing human actions.
Verifiability	Claims must be based on open, accessible evidence.
Falsifiability	It must be possible to prove a claim wrong.
Dialectics	Reality created by contradictory forces coming together.
Periodisation	Dividing history into named time periods (problematic).
Shruti	“That which is heard” – oral, divinely revealed tradition.
Smriti	“That which is remembered” – written tradition.
Lokayata / Charvaka	Ancient Indian materialist/empiricist tradition.
Epistemic Violence	Imposing one knowledge system on another as a form of domination.
Subaltern	Marginalised people (peasants, tribals) whose voice is suppressed.
Oriental Despotism	Colonial claim that all Indian rulers were tyrants.
Drain of Wealth	R. C. Dutt’s theory of economic extraction from India to Britain.
Geographical Determinism	Oversimplified view that geography alone determines history.
Enabling/Limiting Conditions	Geography provides possibilities and constraints, not deterministic outcomes.

8 Key Figures Reference

Figure	Significance
Copernicus & Galileo	Heliocentric revolution; de-centering of creation.
Newton	Absolute laws of nature; God as “First Cause” only.
Carl Sagan	<i>Pale Blue Dot</i> ; de-centering of nature.
Leonardo da Vinci	Renaissance; human form in art.
Herodotus	First historian; “historia” = inquiry.
Thucydides	Evaluated evidence; origin of objective fact.
Ibn Khaldun	Principles to evaluate historical evidence.
Ibn Rushd (Averroes)	Categorised errors in historical investigation.
Al-Khwarizmi	Origin of “algorithm”; intellectual milieu of algebra.
Edward Gibbon	Primary vs secondary sources; secular history.
Leopold von Ranke	“Only facts, no interpretation”; <i>wie es eigentlich gewesen</i> .
E. H. Carr	<i>What is History?</i> ; facts are constituted by historians.
William Jones	Asiatic Society of Bengal; Orientalist approach.
James Mill	<i>History of British India</i> ; Hindu/Muslim/British periodisation.
Macaulay	English education policy; Indian Penal Code.
James Prinsep	Deciphered Brahmi script.
Alexander Cunningham	Founded ASI (1861); Ashoka’s edicts.
Nehru	<i>Discovery of India</i> ; nationalist school.
Romila Thapar	Historian; work on Ashoka.
R. C. Dutt	Economic historian; drain of wealth.
Jadunath Sarkar	Mughal and Maratha period historian.
D. D. Kosambi	Mathematician at TIFR; Marxist history pioneer; iron technology thesis.
R. S. Sharma	Material culture and class structures.
Irfan Habib	Mughal agrarian crisis.
Ranajit Guha	Founded Subaltern Studies collective (1981).
Birsa Munda	Tribal leader; revolt against British; subaltern example.
Rudyard Kipling	“White man’s burden” ideology.

9 Potential Exam Questions & Answer Frameworks

Exam Strategy

- The quiz is **45 minutes** and covers **all 4–5 history lectures + compulsory readings**.
- Content **spoken in class but not on slides** will appear.
- **Creative thinking** and going beyond bullet points is rewarded.
- Use examples from **encouraged readings** (Somnath, *The Name of the Rose*) for extra marks.

9.1 Short Answer / Conceptual Questions

Q1. What is the difference between “the past” and “history”?

The past is amorphous and undifferentiated – everything that happened. History is a specific, human-manufactured narrative with a beginning and end, curated from the past through selection and interpretation.

Q2. Explain the concept of “disenchantment of nature.”

After the heliocentric revolution (Copernicus, Galileo, Newton), nature was understood as governed by fixed, discoverable laws – not by divine whim or miracles. This “disenchanted” nature by removing supernatural explanations.

Q3. Why does the professor say “facts are constituted, not found”?

Following E. H. Carr: facts do not exist independently of the historian. Historians select which events to call “facts,” interpret them, and arrange them into narratives. The same evidence can yield different “facts” depending on the historian’s perspective, questions, and methods.

Q4. What is the problem with the Ancient/Medieval/Modern periodisation?

It carries moral loads (golden age vs dark ages), is Eurocentric (designed for European history), does not fit non-Western timelines, and the boundary of “modern” is arbitrary (1757? 1793? Railways?). Technologies and periods overlap rather than neatly replace each other.

Q5. Compare any two schools of Indian historiography.

E.g., Colonial vs Marxist: Colonial historians argued India had no unity and the British were agents of change. Marxist historians (Kosambi, Irfan Habib) argued that material conditions (iron technology, agrarian systems) drove history, and 1947 did not fundamentally change India’s class structure – merely transferred power from British to Indian elites.

Q6. What is “epistemic violence”?

A concept from Subaltern Studies: imposing Western categories of knowledge on non-Western peoples is itself a form of violence – it delegitimises indigenous knowledge systems and ways of understanding the world.

Q7. Why is the Ganges valley called the “powerhouse of Indian history”?

Its extremely moist alluvial soil produces massive agricultural surplus with minimal effort. This surplus enabled feeding non-agricultural populations (soldiers, builders, scholars), which in turn enabled the building of empires, cities, and civilisations. Only the Yangtze basin in China is comparable.

Q8. Explain the “forest vs. field” dialectic.

Indian history is a history of conflict and complementarity between forest and field. Agriculture requires clearing forests (conflict), but forests provide charcoal for iron-making, grazing, biodiversity (complementarity). This continues today as industrial development vs. forest/tribal land.

Q9. What role did iron technology play in Indian history?

Iron axes enabled deforestation of the Ganges valley → agriculture → surplus → cities and empires. Iron required charcoal (from trees), creating a dialectic: forests were cut to make the iron that

cut more forests. 300–400 years after iron, Buddhism and Jainism emerged in the new social conditions.

Q10. How did the British census change India?

The census (esp. 1881 onward) counted every person regardless of caste/gender/religion. By using religion, caste, and language as primary categories, it made populations self-aware of their numbers, enabling political mobilisation. Even Vande Mataram’s “seven crore voices” comes from census data.

Q11. What is the significance of the Great Trigonometrical Survey?

Started 1802, completed 1872. Applied trigonometry at planetary scale on a curved surface. Created the first reliable map of India. Mt. Everest was calculated from Patna in the 1850s. Before this, India used mandala maps that no one can read today.

Q12. Why does the professor reject “geographical determinism”?

Geography provides enabling and limiting conditions, not deterministic outcomes. The Ganges valley’s potential required human agency: mastery of horse, fire, and iron to convert forest into farmland. Without these technologies, the geography alone would not have produced empires.

Q13. How did mineral coal change Indian history?

Charcoal gave ~125 calories per 100 spent; mineral coal gave ~190. This new energy source enabled the British industrial revolution, railways, modern weapons. The British conquered India not through racial superiority but through superior energy technology.

Q14. What were the contributions of Ibn Khaldun and Ibn Rushd to historical method?

Ibn Khaldun developed principles for evaluating evidence. Ibn Rushd categorised errors in historical investigation: partisanship, unexamined beliefs, insufficient knowledge. Both were part of the Islamic intellectual renaissance that also produced al-Khwarizmi (algorithms, algebra).

Q15. Why is the Kushan Empire significant but overlooked in Indian history?

It lasted ~300 years, spanning from Uzbekistan to UP – 15–17 generations lived under it. But it doesn’t fit the map of modern India, so it is treated as an “aberration” rather than a central fact of Indian history. This shows how we project modern maps onto the past.

9.2 Essay-Type Questions

E1. “History is not the past; it is a human-manufactured narrative.” Discuss with reference to the evolution of historical method from Herodotus to E. H. Carr.

E2. Critically evaluate the four schools of Indian historiography. Which school do you find most persuasive, and why?

E3. “The Ganges valley is a human artifact.” Explain this statement with reference to the role of technology (horse, fire, iron) in shaping Indian history.

E4. How did the British change the historical geography of India? Discuss with reference to railways, the census, and trigonometric surveys.

E5. “Every enabling condition is also a limiting condition.” Discuss with examples from river basins, mountains, and forests in Indian history.

10 Final Revision Checklist

Things You MUST Know

- | | |
|--|--|
| 1. Heliocentric revolution & its significance | 16. D. D. Kosambi & iron technology thesis |
| 2. Disenchantment of nature | 17. Subaltern Studies & epistemic violence |
| 3. Humanism (academic definition) | 18. River basins as historical entities |
| 4. Human sciences vs natural sciences | 19. Indus vs Ganges valley differences |
| 5. E. H. Carr: facts are constituted | 20. Forest–field dialectic |
| 6. Herodotus → Thucydides → Ibn Khaldun → Gibbon → Ranke | 21. Horse → fire → iron sequence |
| 7. Verifiability and falsifiability | 22. Deforestation of Ganges = human artifact |
| 8. History ≠ Past | 23. Energy: charcoal vs mineral coal |
| 9. Cyclical vs linear time | 24. Census timeline & impact |
| 10. Stone Ages & Metal Ages (names + dates) | 25. Great Trigonometrical Survey |
| 11. Why periodisation is problematic | 26. Kushan & Chola empires as “aberrations” |
| 12. Four schools of Indian historiography | 27. Mandala maps vs modern maps |
| 13. James Mill’s periodisation | 28. Geographical determinism critique |
| 14. Macaulay’s education policy | 29. Birsa Munda as example |
| 15. Prinsep / Cunningham / ASI / IVC | 30. Lokayata / Charvaka tradition |

Last-Minute Tips

- Read **Chapters 1–3 of E. H. Carr** – questions *will* come from it.
- Read the **Silver Blaze** Sherlock Holmes story – the detective analogy recurs.
- The professor said: “some of it which is **not on the slides but I have spoken** will be there in the question paper.”
- Show **creative thinking** and your own analysis – don’t just reproduce bullet points.
- Use **specific examples** (Jallianwala Bagh, Birsa Munda, Kushan Empire, census data in *Vande Mataram*).

Good luck on your exam!